

Technical University of Munich

School of Computation, Information and Technology
-- Informatics --

Bachelor's Thesis in Informatics

LLM Based Formative Feedback for Theses

LLM-basiertes formatives Feedback für Abschlussarbeiten

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Abstract

Supervisors of theses spend considerable time on non-content-related feedback such as document structure, coherence, and linguistic accuracy. Much of this feedback is repetitive across students and disciplines, draining supervisors and leaving less capacity for substantive research discussions. For students, it translates into slower feedback turnaround.

This thesis explores how large language models (LLMs) can generate formative feedback on theses and proposals based on supervisor-provided writing guidelines. Beyond a single draft, the tool also compares iterations to highlight improvements and surface persistent issues with more explanatory feedback. We evaluate single-prompt, prompt-chaining, and agentic setups, and assess trade-offs between local and cloud hosting. The tool integrates into Thesis Management, the thesis-lifecycle portal used by around 50 research groups at the Technical University of Munich, so supervisors can generate, revise, and publish feedback within their workflow.

1 Introduction

Bachelor's and Master's theses represent important final projects that students complete before graduating [TP21]. Under the guidance of a supervisor, students conduct independent research and engage in depth with their chosen topic, applying the knowledge they have built throughout their studies to the academic world [AAM17]. The goal of a thesis is for students to demonstrate the skills and academic competences they acquired during their studies and to establish themselves as members of the research community [AAM17, TP21]. Before starting their research project, students typically write a research proposal. In this proposal, they define the research question and problem they want to explore, outline their methodology, and specify the scope of their work.

At around 50 research groups at the Technical University of Munich (TUM), supervisors and students manage the thesis lifecycle through Thesis Management[1], an in-house portal developed at TUM. Through this portal, they coordinate proposal submissions, feedback exchanges, and the final thesis hand-in. The proposal stage follows an iterative workflow: the student submits a draft, the supervisor enters feedback as actionable items, and the student addresses each item, ticks it off, and uploads a new version. This cycle continues until the proposal is accepted. The student, by contrast, hands in the thesis itself as a single final submission.

2 Problem

For students' research proposals and theses, supervisors often provide feedback that is critical to their progress. The supervisor takes the role of an experienced peer and coach in a mentoring environment, gradually increasing the student's level of independence. Theses are often open-ended assignments, so feedback frequently becomes the only form of instruction students receive [BBE10].

Supervisors provide feedback on a wide range of topics, including the content itself, the structure of the document and its chapters, writing coherence and cohesion, and linguistic accuracy. Providing this feedback is time-intensive, and especially for non-content-specific categories, highly repetitive: because thesis students are still developing their academic skills, supervisors tend to give similar feedback across students, disciplines, and language backgrounds [BBE10]. While some supervisors are willing to help students with proofreading, this is not their primary task [BBE10].

Giving repetitive feedback is a draining activity for supervisors [Bog+07]. This may lead to reduced concentration, causing supervisors to overlook actual content or conceptual mistakes. Moreover, the time spent on repetitive feedback leaves less time for what matters most: helping students with their research and providing feedback on substantive topics.

[1] https://github.com/ls1intum/thesis-management

While the problem is less severe for students, they still experience its consequences. When supervisors find the feedback task tedious, they may take longer to respond, especially when it involves repetitive, non-content-related issues. Each university often defines its own guidelines for academic writing and document structure, and the academic community has not standardized these conventions. Having to wait for feedback on purely formal aspects of a thesis or proposal can be frustrating for students, as these issues are governed by institutional conventions rather than research substance.

3 Motivation

Integrating large language models (LLMs) to assist supervisors with non-content-related feedback could improve the thesis writing process. As Figure 1 illustrates, an LLM-powered `Reviewer` works alongside the supervisor, evaluating the thesis against the research group's writing guidelines and generating automatic formative feedback [NM06]. This `Reviewer` covers exactly the repetitive, non-content-related layer of feedback that currently drains supervisors, freeing them to focus on specific content and on discussions with their students, which is key in mentoring environments [BBE10, NM06]. The quality of feedback improves as supervisors have more time to engage deeply with the thesis, leading to better outcomes overall. Additionally, supervisors can provide feedback faster, making the process more scalable.

Better and faster feedback has significant advantages for students as well. When feedback arrives sooner, students can act on it while the writing process is still fresh in their minds. The student brings both knowledge, comprising theoretical and practical domain expertise, and beliefs about how to approach the task and how to work and write scientifically, which include the student's personal interpretation of the writing guidelines. Drawing on this knowledge and these beliefs, the student writes the thesis. Through self-regulated learning, the student engages with the `Reviewer`'s feedback, reflecting on it independently or

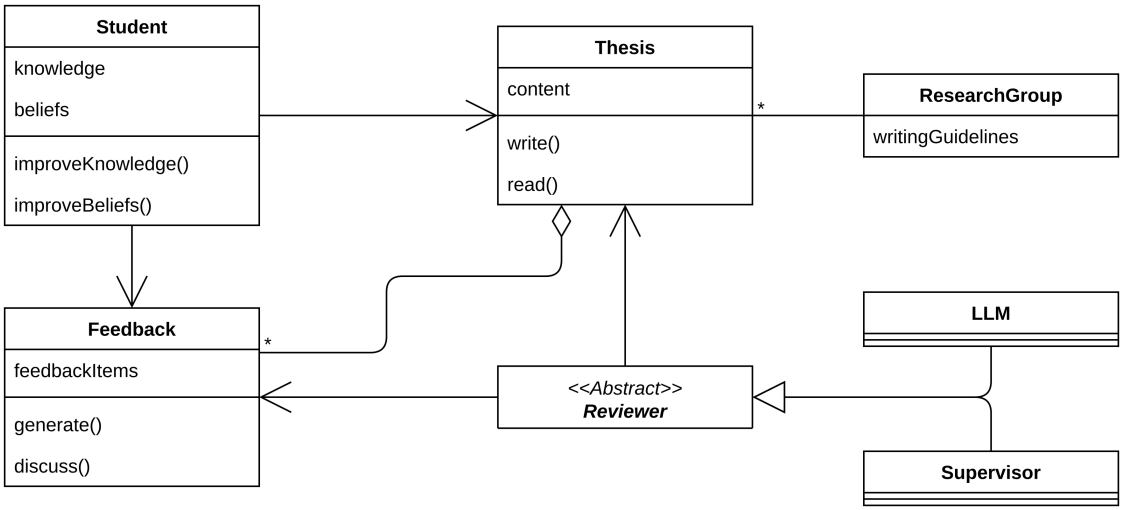


Figure 1: Analysis object model of the envisioned feedback process: an LLM `Reviewer` works alongside the supervisor, generating formative feedback on the thesis that the student uses to improve their knowledge and beliefs through self-regulated learning [NM06].

discussing it with the supervisor, and uses it to improve their knowledge, beliefs, and broader strategy for proceeding with the work [NM06]. Because the feedback tool is powered by LLMs, students can obtain instant formative feedback at any point during the writing process.

4 Objective

The scope of the thesis is to fulfill the following goals:

1. Generate Feedback Based on Guidelines
2. Highlight Changes Across Iterations
3. Compare Prompting Techniques and LLM Models
4. Integrate With Thesis Management

4.1 Generate Feedback Based on Guidelines

Supervisors should be able to provide their own scientific writing guidelines as well as conventions on the style of theses and proposals. Based on these guidelines, the LLM can provide formative feedback that is restricted to the aspects they cover. Since guidelines may not already be in a suitable format, the LLM may first

need to preprocess them into a structured representation it can act on. Ethically, it is difficult to let LLMs grade a final thesis, as this constitutes normative feedback and requires close scrutiny. Formative feedback, on the other hand, is less critical since it serves as input for students to improve their work. The idea is that students interpret the feedback themselves and discuss it, either independently or with their supervisor. Therefore, it is less important that every output of the LLM is perfectly correct.

This approach supports several modes of feedback generation. In one mode, no LLM-generated feedback is produced at all and all feedback comes from the supervisor. Alternatively, the student can obtain instant feedback on their proposal or thesis directly from the LLM, while keeping in mind that the supervisor has not reviewed this output. A third mode combines both sources, as illustrated in Figure 2. In this combined mode, multiple specialized agents analyze specific parts of the thesis in parallel. The supervisor then aggregates and reviews the resulting feedback and adds their own feedback. Finally, the supervisor merges both sets and publishes them to the student.

4.2 Highlight Changes Across Iterations

When the feedback tool has access to previous iterations of the proposal or thesis, it can compare the latest submission with earlier ones, both at the level of the text itself and at the level of feedback items raised previously. The goal is to detect improvements and acknowledge them in the new feedback, for example by explicitly mentioning items the student has resolved. Calling out these aspects encourages students in the areas they have successfully addressed.

If a regression occurs or an issue persists across iterations, the previous feedback was likely hard for the student to understand and act on. A regression means an aspect that was previously fine has worsened, while a persistent issue means earlier feedback flagged the same problem and the student has not yet resolved it. In such cases, the feedback tool should adapt its style, for example by being more explanatory, more verbose, or providing concrete examples that illustrate

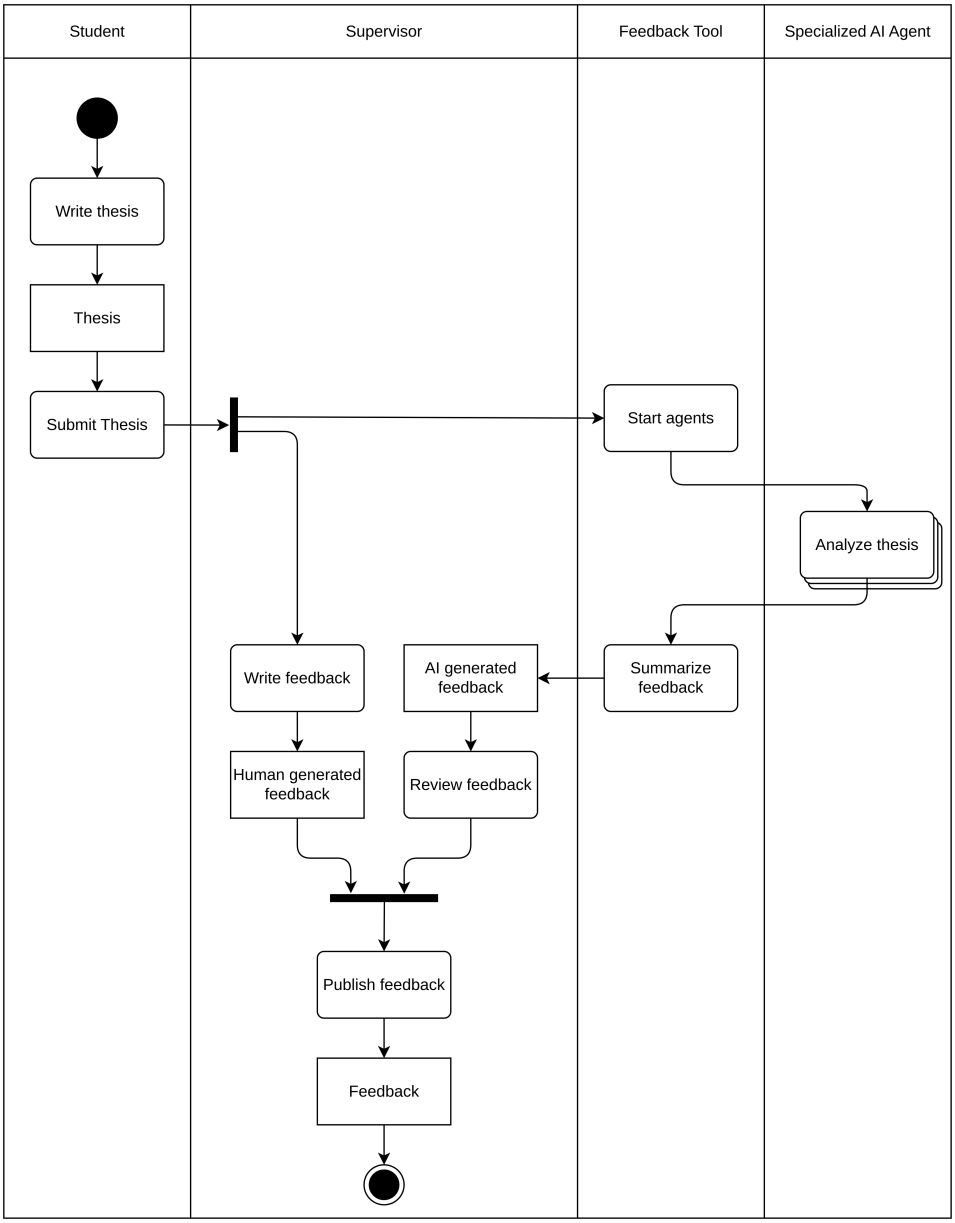


Figure 2: Activity diagram of the envisioned feedback workflow. After the student submits a thesis, the supervisor and the feedback tool work in parallel, with specialized agents analyzing the thesis. The supervisor then revises the AI-generated draft, joins it with their own feedback, and publishes the result to the student.

the desired outcome. Where the feedback tool repeatedly fails to communicate an issue, it should defer the discussion to the supervisor.

4.3 Compare Prompting Techniques and LLM Models

Based on the writing guidelines, the feedback tool needs to cover a range of aspects and topics across the thesis, from document structure and chapter organization to writing coherence, cohesion, and linguistic accuracy. A single plain prompt to an LLM may not cover all of them reliably, since attention can dilute when many heterogeneous criteria are bundled into one request. Alternatives include prompt chaining and agentic setups, in which the LLM itself identifies where further lookups into the proposal are required and triggers specialized feedback generations focused on a narrower set of aspects. We benchmark which of these options performs best in this setting.

In addition, privacy regulations and cost require careful consideration of when to use locally self-hosted LLMs versus cloud-hosted ones. Self-hosted models keep student drafts inside the university's infrastructure but are limited by the available hardware, while cloud-hosted models offer stronger capabilities at usage-based prices. We compare the available providers and decide which model to use where, based on the resulting constraints.

4.4 Integrate With Thesis Management

Around 50 research groups at TUM use Thesis Management to manage the thesis lifecycle. Within this portal, supervisors can record feedback that students can review and resolve in subsequent proposal iterations. The portal already models the concepts of feedback and proposal submissions and exposes them through a user interface, so we integrate the feedback tool into this existing system. With a single click, supervisors invoke the feedback tool to write feedback directly into Thesis Management. They can then edit or extend the generated feedback before sharing it with the student.

By building on existing processes, the integration causes the least disruption to supervisors and delivers the greatest simplification of their workflow. At present, the feedback flow in Thesis Management only exists for the proposal, since the

thesis is handed in as a single final submission without an iterative feedback cycle. We extend the existing proposal feedback flow to also cover the thesis, enabling formative feedback during the writing of the thesis itself.

5 Schedule

The Bachelor thesis starts on 26.06.2026 and the submission is on 26.10.2026.

We divided the time into several two-week iterations, each containing tasks to fulfill the overall goals:

Iteration 1 (08.06.2026 - 2 weeks)

- Modularize Thesis Management to prepare integration of the feedback module (Objective 4)

Iteration 2 (22.06.2026 - 2 weeks)

- Implement core feedback generation pipeline (Objective 1)
- Define input/output contract and structured output format (Objective 1)

Iteration 3 (06.07.2026 - 2 weeks)

- Connect feedback tool to Thesis Management (Objective 4)
- Extend feedback workflow in Thesis Management for thesis (Objective 4)

Iteration 4 (20.07.2026 - 2 weeks)

- Allow supervisors to upload custom guidelines (Objective 1)
- Preprocess guidelines into a structured representation usable by the feedback tool (Objective 1)

Iteration 5 (03.08.2026 - 2 weeks)

- Build prototype implementations for agentic and prompt chaining setups (Objective 3)
- Evaluate agentic and prompt chaining setups (Objective 3)

Iteration 6 (17.08.2026 - 2 weeks)

- Build prototype implementations for the selected LLM models and providers (Objective 3)
- Evaluate different LLM models and providers (Objective 3)

Iteration 7 (31.08.2026 - 2 weeks)

- Detect issues that persist across proposal iterations (Objective 2)
- Adapt feedback style for persistent issues (more verbose, or defer to supervisor) (Objective 2)

Iteration 8 (14.09.2026 - 3 weeks)

- Detect issues addressed across proposal iterations (Objective 2)
- Surface acknowledgements for improvements in the feedback output (Objective 2)

Iteration 9 (05.10.2026 - 3 weeks)

- Review the system and check for bugs
- Documentation

Bibliography

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Transparency in the Use of AI Tools

In preparing this proposal, I made use of several AI-based tools across different stages of the writing process. For grammar and style correction, I used Grammarly to identify grammatical errors, improve sentence structure, and refine the overall writing style of my drafts. For translation and language enhancement, I used DeepL to translate passages between German and English and to improve the fluency of phrasings. For content generation and idea expansion, I used ChatGPT, OpenAI Codex, Claude, and Claude Code to reformulate sentences I had already drafted, expand on ideas I had outlined, and obtain suggestions for clearer phrasing. For coding assistance, I used GitHub Copilot, ChatGPT, OpenAI Codex, Claude, and Claude Code to support Typst formatting, generate snippets for figures and macros, and resolve template-related issues. For citation assistance, I used Zotero to manage my references and to ensure proper citation formatting throughout the proposal. I have carefully checked all texts created with these tools to ensure that they are correct and make sense.